ELECTRICAL LEGEND

SYMBOL	DESCRIPTION
	OVERLOAD RELAY HEATER
M	MOTOR, SQUIRREL CAGE INDUCTION
DOL	DIRECT ON-LINE (DOL) STARTER
DOL	REVERSE DIRECT ON-LINE (RDOL) STARTER
VFD	VARIABLE FREQUENCY DRIVE (VFD)
SS	SOFT STARTER
S/D	STAR-DELTA STARTER
	SINGLE PHASE MOULDED CASE CIRCUIT BREAKER
	MOULDED CASE CIRCUIT BREAKER
	AIR CIRCUIT BREAKER
	RESIDUAL CURRENT OPERATED CIRCUIT BREAKER (RCBO)
	DRAWOUT FUSE
	DRAWOUT FUSED SWITCH
	SURGE ARRESTER
10	CAPACITOR - KVAR INDICATED
VS V 0-600V	METER WITH SWITCH - SCALE RANGE SHOWN
	CONTACTOR
M	MOTOR OPERATED DISCONNECTOR SWITCH
	DATA LINK
	DISCONNECTOR
	WITHDRAWABLE MCCB
	WITHDRAWABLE ACB
	SF6 CIRCUIT BREAKER (MOTORISED)
	DISCONNECTOR C/W EARTHING SWITCH (MOTORISED)
	FUSE
	VOLTAGE INDICATOR
	MAIN DISTRIBUTION BOARD/ DISTRIBUTION BOARD

SYMBOL	DESCRIPTION
(3)	CURRENT TRANSFORMER, NUMBER INDICATED
(1)	ZERO SEQUENCE CURRENT TRANSFORMER, NUMBER INDICATED
	TRANSFORMER, WINDING CONNECTION INDICATED, PRIMARY AND SECONDARY VOLTAGES, PHASE AND I-RATING INDICATED ON SINGLE-LINE DIAGRAM
	AUTOMATIC TRANSFER SWITCH
K2	KEYED INTERLOCK
△ NER OR NER	NEUTRAL EARTHING RESISTOR
	ISOLATOR
	TRANSFORMER WITH TERTIARY WINDING, WINDING CONNECTION INDICATED, PRIMARY, SECONDARY AND TERTIARY VOLTAGES, PHASE AND MVA RATING INDICATED ON SINGLE-LINE DIAGRAM
OR	DRAWOUT SYMBOL
N E ATS	AUTOMATIC TRANSFER SWITCH
	MULTI-WINDING ISOLATION TRANSFORMER FOR VFD APPLICATION (24-PULSE)

ABBREVIATIONS

ABBREVIATIONS	DESCRIPTION
A	AMMETER, AMPERE, AMBER
ACB	AIR CIRCUIT BREAKER
AF	AMPERE FRAME
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AS	AMMETER SWITCH, AMPERE SENSOR
ASC	ANALOGUE SIGNAL CABLE
ASU	AIR SUPPLY UNIT
AT	AMPERE TRIP
ATC	AUTOMATIC THROWOVER CONTROL
ATS	AUTOMATIC TRANSFER SWITCH
BAT	BATTERY
BC	BARE COPPER
BCH	BATTERY CHARGER
BRKR	BREAKER
BT	TELEPHONE TERMINATION BOX
BW	BUSWAY
C	CONDUIT, CONTACTOR, CLOSE
CB	CIRCUIT BREAKER
CC	CONTROL CABLE
CL	CLASS
CCL	CONTROL CIRCUIT CABLE LADDER
CCR	CABLE CHAMBER
CCT	CONTROL CIRCUIT CABLE TRAY
CCTV	CLOSED CIRCUIT TELEVISION
CKT	CIRCUIT
CLG	CEILING
COND	CONDUIT
CPC	CIRCUIT PROTECTION CONDUCTOR
CPT	CONTROL POWER TRANSFORMER
CR	CONTROL RELAY
CRE	CORROSION-RESISTANT
CS	CONTROL SWITCH, CONTROL STATION
CT	CURRENT TRANSFORMER
C/T	CLOSE/TRIP
CU	CONSUMER UNIT
C/W	COMPLETE WITH
DA	DISCRETE ANALOG
DB	DISTRIBUTION BOARD
DBE	DISTRIBUTION BOARD EMERGENCY
DC	DIRECT CURRENT
DCP	DC PANEL
DIV	DIVISION
DOL	DIRECT ON-LINE STARTER
DP	DRAW PIT
DSC	DISCONNECT
DSW	DISCONNECTOR SWITCH
E	EARTH, EMPTY
ECPS	ELECTTRONIC CONTROL AND PROTECTION SYSTEM
EF	EARTH FAULT RELAY
ELR	EARTH LEAKAGE RELAY
EM	ELECTRONIC POWER METER
EMG	EMERGENCY POWER GENERATOR
EO	ELECTRIC OPERATOR
EQPT	EQUIPMENT
ES	EARTHING SWITCH
ESS	EMERGENCY SHUTDOWN SWITCH EMERGENCY STOP SWITCH
ETM	ELAPSED TIME METER
EXST	EXISTING
FDR	FEEDER
F	FUSE
FLR	FLOOR
FLUOR	FLUORESCENT
FPP	FIBRE PATCH PANEL
FR	FIRE RESISTANT
FTG	FITTING
FVNR	FULL VOLTAGE NON-REVERSING
FVR	FULL VOLTAGE REVERSING
G	GREEN, GENERATOR
GALV	GALVANISED
GEN	GENERATOR
GRS	GALVANISED RIGID STEEL CONDUIT
GIS	GAS-INSULATED SWITCHGEAR

ABBREVIATIONS	DESCRIPTION
H	HIGH SPEED
HID	HIGH INTENSITY DISCHARGE
HPS	HIGH PRESSURE SODIUM
HS	HAND SWITCH
HV	HIGH VOLTAGE
HVDS	HIGH VOLTAGE DISCONNECT SWITCH
HVL	HIGH VOLTAGE CABLE LADDER
HVPC	HIGH VOLTAGE POWER CABLE
HVSG	HIGH VOLTAGE SWITCHGEAR (ABOVE 1000 VOLTS)
IC	INTERRUPTING CAPACITY INSTRUMENT CABLE
I &C	INSTRUMENTATION AND CONTROL
IDF	INTERMEDIATE TELEPHONE DISTRIBUTION FRAME
I/L	INTERLOCK
INCAND	INCANDESCENT
INST	INSTANTANEOUS
ISOL	ISOLATOR
IST	INSTRUMENT SYSTEM CABLE TRAY
ISW	ISOLATOR SWITCH
ITR	ISOLATION TRANSFORMER
J, JB	JUNCTION BOX
JT	JOINT OR SPLICE
K	KEY INTERLOCK
kV	KILOVOLT
kA	KILO AMPERES
kVA	KILOVOLT AMPERES
kWH	KILOWATT HOUR
L	LIGHTING CONTACTOR, LOW SPEED
LCP	LOCAL CONTROL PANEL
LCS	LOCAL CONTROL STATION
LMCP	LOCAL MOTOR CONTROL PANEL
LOS	LOCKOUT STOP PUSHBUTTON
LR	LATCHING RELAY
LT FLEX	LIQUIDTIGHT FLEX CONDUIT
LTG	LIGHTING
LTS	LIGHTS
LV	LOW VOLTAGE
LVL	LOW VOLTAGE CABLE LADDER
LVT	LOW VOLTAGE CABLE TRAY
LVTk	LOW VOLTAGE CABLE TRUNKING
LVPC	LOW VOLTAGE POWER CABLE
LVSG	LOW VOLTAGE SWITCHGEAR (BELOW 1000 VOLTS)
M	MAGNETIC CONTACTOR COIL
MB	MAIN BREAKER
MCB	MINIATURE CIRCUIT BREAKER
MCC	MOTOR CONTROL CENTRE
MCCB	MOULDED CASE CIRCUIT BREAKER
MDF	MAIN TELEPHONE DISTRIBUTION FRAME
MH	METAL HALIDE
MO	MOTOR OPERATOR
MPR	MULTI-FUNCTION PROTECTIVE RELAY
MS	MOTOR STARTER
MT	MOUNT
MTD	MOUNTED
MW	MEGAWATT
MVA	MEGAVOLT AMPERE
N	NEUTRAL
NA	NON-AUTOMATIC
NC	NORMALLY CLOSED
NER	NEUTRAL EARTHING RESISTOR
NL	NIGHT LIGHT
N.O.	NORMALLY OPEN
NP	NAMEPLATE
NPP	NETWORK PATCH PANEL
OC	ON CENTRE, OPEN/CLOSE
O/C	OVERCURRENT RELAY
ONAF	OIL NATURAL, AIR FORCED
ONAN	OIL NATURAL, AIR NATURAL
OL	OVERLOAD RELAY

ABBREVIATIONS	DESCRIPTION
P	PHASE
P	PUMP, POWER
PADP	PUBLIC ADDRESS DISTRIBUTION PANEL
PB	PULL BOX, PUSHBUTTON SWITCH
PC	PHOTOCELL
Pf	PULSE FREQUENCY
PH	PHASE
PMR	PHASE MONITOR RELAY
PMS	POWER MONITORING SYSTEM
PNL	PANEL
PR	PROTECTIVE RELAY AND POWER MONITORING PANEL
PS	PRESSURE SWITCH
PT	POTENTIAL TRANSFORMER
PVC	POLYVINYL CHLORIDE
PW	PILOT WIRE
P/W	PW MAIN PROTECTION RELAY
R	RED
RCCB	RESIDUAL CURRENT CIRCUIT BREAKER
RCPT	RECEPTACLE
REQD	REQUIRED
RM	REVENUE METERS
RMU	RING MAIN UNIT
RS	RIGID STEEL CONDUIT
RVNR	REDUCED VOLTAGE NON-REVERSING
RVR	REDUCED VOLTAGE REVERSING
SA	SURGE ARRESTOR
SCCR OR SCR	SHORT CIRCUIT CURRENT RATING
SD	STAR-DELTA STARTER
S/N	SOLID NEUTRAL
SPD	SPEED
SPN	SINGLE PHASE AND NEUTRAL
SSO	SWITCHED SOCKET OUTLET
SSS	SOFT START STARTER
SST	STAINLESS STEEL
S/T	SHUNT TRIP
SV	SOLENOID VALVE
SW	SWITCH
SWBD	SWITCHBOARD
SWGR	SWITCHGEAR
T	THERMOSTAT, TRIP
TB	TERMINAL BOARD
TD	TEMPERATURE DETECTOR RELAY
TDP	TELEPHONE DISTRIBUTION PANEL
TC	TIME CLOSE
TDR	TIME DELAY RELAY
TJB	TERMINAL JUNCTION BOX
T.O.	TIME OPEN
TPN	THIRD PHASE AND NEUTRAL
TRF	TRANSFORMER
TS	TEMPERATURE SWITCH
TSP	TWISTED SHIELDED PAIR
TST	TWISTED SHIELDED TRIAD
TYP	TYPICAL
UH	UNIT HEATER
UPS	UNINTERRUPTIBLE POWER SUPPLY
UVR	UNDER VOLTAGE RELAY
V	VOLTMETER, VOLT, VOLTAGE
VCB	VACUUM CIRCUIT BREAKER
VFD	VARIABLE FREQUENCY DRIVE
VS	VOLTMETER SWITCH
VT	VOLTMETER TRANSFORMER
W	WATT / WIRE
WH	WATER HEATER
WHD	WATTHOUR DEMAND METER
WHX	WATT TRANSDUCER
WP	WEATHERPROOF
WHX	WATT TRANSDUCER
XFMR	TRANSFORMER
XLT/XLVT	EXTRA LOW VOLTAGE CABLE TRAY
XLTk/XLVTk	EXTRA LOW VOLTAGE CABLE TRUNKING

TENDER

REV	DATE	DESCRIPTION	REVIEWED	AUTHORISED
T1	20/05/22	ISSUED FOR TENDER	LBL	LF

CLIENT:
PUB SINGAPORE'S NATIONAL WATER AGENCY

PROGRAMME MANAGER:
binnies AECOM
A JOINT VENTURE DELIVERING THE DTSS2 PROJECT

CONSULTANT:
Jacobs
CH2M HILL Singapore Pte Ltd

PROJECT TITLE:
TUAS WATER RECLAMATION PLANT (TWRP)
DTSS PHASE 2

DRAWING TITLE
ELECTRICAL
LEGEND & ABBREVIATIONS

DRAFTED: UM	DESIGNED: GR	SHEET SIZE: A1
DRAFTING CHECK: DW	DESIGN CHECK: PC/SK	SCALE: NA
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